Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech *Time: 03 Hours*
Code: 105401 *Subject: Computer Organization and Architecture* *Full Marks: 70*

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The unit responsible for fetching, decoding, and executing instructions is
(i) ALU (iii) Memory Unit
(ii) Control Unit (iv) I/O Unit

(b) Which addressing mode uses the contents of a register to determine the effective address?
(i) Signed integers (iii) Direct addressing
(ii) Indirect addressing (iv) Register addressing

(c) Two's complement representation is primarily used for
(i) Signed integers (iii) Instruction opcodes
(ii) ASCII characters (iv) Floating point numbers

(d) Booth's algorithm is used for
(i) Floating point operations (iii) Character coding
(ii) Division (iv) Multiplication

(e) In floating point representation, the part representing significant digits is called
(i) Mantissa (iii) Sign bit
(ii) Exponent (iv) Bias

(f) x86 architecture belongs to which processor family?
(i) RISC (iii) CISC
(ii) EPIC (iv) ARM

(g) Memory chips used in main memory are typically
(i) Optical memory (iii) Flash
(ii) Magnetic tapes (iv) DRAM

(h) USB is an example of
(i) Networking device (iii) Storage device only
(ii) Standard I/O interface (iv) Wireless protocol

(i) Pipeline speedup ideally increases with
(i) Number of control units (iii) Size of RAM
(ii) Number of pipeline stages (iv) Size of cache

(j) In cache memory, FIFO and LRU are examples of
(i) Mapping techniques (iii) Addressing techniques
(ii) Replacement algorithms (iv) Write policies

Q.2 (a) Explain the functional blocks of a computer with neat diagrams: CPU, Memory, and I/O subsystems. **[7]**

(b) Explain addressing modes with examples. Compare at least five different addressing techniques. **[7]**

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Q.3	(a)	Explain all types of number representations: signed integers, fixed point, floating point, and character codes.	[7]
	(b)	Describe floating point arithmetic and explain overflow, underflow, normalization, and rounding.	[7]
Q.4	(a)	Describe shift-and-add multiplication and Booth's multiplication algorithm with examples.	[7]
	(b)	Explain the architecture of x86 processors. Discuss its registers, addressing modes, and instruction format..	[7]
Q.5	(a)	Describe the design of a simple hypothetical CPU including instruction set, datapath, and control design.	[7]
	(b)	Explain semiconductor memory organization with diagrams.	[7]
Q.6	(a)	Explain I/O transfer techniques: program-controlled I/O, interrupt-driven I/O, and DMA.	[7]
	(b)	Describe different types of interrupts: hardware, software, vectored, non-vectored, and exceptions.	[7]
Q.7	(a)	Explain the basic concepts of pipelining, including speedup and throughput. Discuss pipeline hazards.	[7]
	(b)	Describe parallel processors and the problem of cache coherency in shared memory systems.	[7]
Q.8	(a)	Explain cache memory organization. Discuss mapping techniques (direct, associative, set-associative), replacement algorithms, and write policies.	[7]
	(b)	Explain signed number representation using sign-magnitude, 1's complement, and 2's complement. Include advantages and disadvantages of each.	[7]
Q.9	(a)	Explain the organization of semiconductor memories such as SRAM and DRAM. Discuss refresh cycles, access times, and their applications.	[7]
	(b)	Discuss memory hierarchy design principles. Explain how access time, cost, and size influence memory organization.	[7]

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